

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

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1. TITLE

Styling Webpages with CSS3 Visuals, Transitions, Box Model, and HTML5 Canvas 2D Graphics

2. AIM

To design and develop an interactive, styled webpage demonstrating CSS3 features (box model, linear gradients, box-shadows, transitions, flexbox and grid layouts) and rendering 2D graphical shapes (rectangle, circle, line, and styled text) on an HTML5 canvas element using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply the CSS3 Box Model (margins, borders, padding, and content).
- To enhance visual aesthetics using linear gradients, rounded corners (border-radius), and box-shadows.
- To build responsive multi-column layouts using CSS Grid and Flexbox.
- To implement interactive hover effects and smooth animations using CSS3 transitions and transforms.
- To utilize the HTML5 <canvas> API with JavaScript 2D rendering context (getContext('2d')).
- To programmatically draw geometric shapes (rectangles, circles, lines) and render styled text dynamically on canvas.

4. THEORY

CSS3 Box Model — Every HTML element is modeled as a rectangular box comprising content, padding (spacing inside the border), border (outline surrounding padding), and margin (spacing outside the border). Setting box-sizing: border-box ensures padding and border dimensions are included within the element's total declared width.

Visual Enhancements & Gradients — CSS3 linear gradients generate seamless multi-hue color progressions without raster image overhead. The box-shadow property produces depth and elevation through horizontal/vertical offsets, blur radius, spread, and semi-transparent RGBA color values.

Interactive Transitions & Transforms — The transition property interpolates CSS property changes smoothly over a specified duration. Combining transitions with transform: translateY(-5px) or scale(1.05) delivers tactile, elevated user feedback upon hover.

HTML5 Canvas 2D Graphics API — The <canvas> element provides a scriptable resolution-dependent bitmap canvas. Accessing its 2D rendering context via getContext('2d') exposes coordinate-based drawing primitives: fillRect()/strokeRect() for rectangular geometry, beginPath(), arc(), and fill() for circular geometry, moveTo()/lineTo() for path vectors, and fillText() for rendering typography.

5. ALGORITHM / PROCEDURE

1. Create two source files: My-Styled-Webpage.html and grey.css.
2. In My-Styled-Webpage.html, structure the layout using semantic <header>, <nav>, <main>, <section>, and <footer> tags.
3. Link the external stylesheet grey.css within the <head> section.
4. Inside <main>, add a canvas section with an HTML5 <canvas id="myCanvas" width="500" height="250"> element.
5. Below the canvas, create a cards container featuring three content cards representing the Box Model, Visual Enhancements, and Interactive Transitions.
6. In grey.css, configure a global reset, set a soft grey gradient body background (linear-gradient(135deg, #f1f5f9, #e2e8f0)), and style the header with charcoal gradients and text shadows.
7. Style the navigation bar with flexbox and configure hover highlights on navigation links.
8. Apply CSS Grid to the cards container (grid-template-columns: repeat(auto-fit, minmax(300px, 1fr))) and style cards with subtle shadows, rounded borders, and translateY hover transitions.
9. In an embedded <script> block, select myCanvas and obtain its 2D rendering context with getContext('2d').
10. Draw a filled blue rectangle with a dark navy border using fillRect(40, 40, 120, 80) and strokeRect(40, 40, 120, 80).
11. Draw a filled red circle with a dark red border using arc(250, 80, 45, 0, 2*Math.PI), fill(), and stroke().
12. Draw a thick green line using moveTo(350, 40),.lineTo(460, 120), and stroke().
13. Render the student name 'Abhishek' on the canvas using fillText() formatted with bold 24px Segoe UI font and center alignment.
14. Open My-Styled-Webpage.html in a web browser and verify all CSS3 visual effects and canvas 2D graphics.

6. PROGRAM

My-Styled-Webpage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <!-- Link to external CSS file -->
  <link rel="stylesheet" href="grey.css">
</head>
<body>

  <!-- Header -->
  <header>
    <h1>My Styled Webpage</h1>
    <p>Demonstrating CSS3 Styling Concepts & HTML5 Canvas Graphics</p>
  </header>

  <!-- Navigation Bar -->
  <nav>
    <ul>
      <li><a href="#canvas">Canvas Graphics</a></li>
      <li><a href="#cards">Content Cards</a></li>
      <li><a href="#about">About Project</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>

  <!-- Main Content Area -->
  <main>
```

```

<!-- Part A: HTML5 Canvas Section -->
<section id="canvas" class="canvas-section">
  <h2>HTML5 Canvas Visuals</h2>
  <canvas id="myCanvas" width="500" height="250">
    Your browser does not support the HTML5 canvas tag.
  </canvas>
</section>

<!-- Part B: Three Content Cards -->
<section id="cards">
  <h2 class="section-title">Featured Concepts</h2>
  <div class="cards-container">

    <!-- Card 1 -->
    <div class="card">
      <h3>CSS3 Box Model</h3>
      <p>The CSS box model is a container that wraps around every HTML element. It consists of
margins, borders, padding, and the actual content.</p>
      <button class="card-btn">Learn More</button>
    </div>

    <!-- Card 2 -->
    <div class="card">
      <h3>Visual Enhancements</h3>
      <p>CSS3 introduces powerful visual capabilities such as linear/radial gradients, rounded
corners, box shadows, and text shadow properties.</p>
      <button class="card-btn">Explore Visuals</button>
    </div>

    <!-- Card 3 -->
    <div class="card">
      <h3>Interactive Transitions</h3>
      <p>Hover effects provide dynamic user feedback by smoothly animating colors, scaling
elements, or adjusting shadows when interacted with.</p>
      <button class="card-btn">View Effects</button>
    </div>

  </div>
</section>

</main>

<!-- Footer -->
<footer>
  <p>&copy; 2026 My Styled Webpage | Designed with HTML5 & CSS3</p>
</footer>

<!-- Part A: JavaScript for Canvas Drawing -->
<script>
  const canvas = document.getElementById("myCanvas");
  const ctx = canvas.getContext("2d");

  // 1. Draw One Rectangle
  ctx.fillStyle = "#3b82f6";
  ctx.fillRect(40, 40, 120, 80);
  ctx.strokeStyle = "#1e3c72";
  ctx.lineWidth = 3;
  ctx.strokeRect(40, 40, 120, 80);

  // 2. Draw One Circle
  ctx.beginPath();
  ctx.arc(250, 80, 45, 0, 2 * Math.PI);
  ctx.fillStyle = "#ef4444";

```

```

        ctx.fill();
        ctx.lineWidth = 3;
        ctx.strokeStyle = "#991b1b";
        ctx.stroke();

        // 3. Draw One Line
        ctx.beginPath();
        ctx.moveTo(350, 40);
        ctx.lineTo(460, 120);
        ctx.lineWidth = 5;
        ctx.strokeStyle = "#10b981";
        ctx.stroke();

        // 4. Display Name as Text
        ctx.font = "bold 24px 'Segoe UI', sans-serif";
        ctx.fillStyle = "#0f172a";
        ctx.textAlign = "center";
        ctx.fillText("Abhishek", 250, 200);
    </script>
</body>
</html>

```

grey.css

```

/* CSS Reset & Global Styling */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    /* Soft grey linear gradient background */
    background: linear-gradient(135deg, #f1f5f9, #e2e8f0);
    color: #1e293b;
    line-height: 1.6;
    min-height: 100vh;
    display: flex;
    flex-direction: column;
}

/* Header Styling - Dark Charcoal to Slate Grey Gradient */
header {
    background-color: #334155;
    background: linear-gradient(to right, #1e293b, #475569);
    color: #ffffff;
    text-align: center;
    padding: 2rem 1rem;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.15);
}

header h1 {
    font-size: 2.5rem;
    font-family: 'Georgia', serif;
    text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.4);
}

header p {
    font-size: 1.1rem;
    margin-top: 0.5rem;
}

```

```

        color: #cbd5e1;
    }

    /* Navigation Bar - Deep Graphite Grey */
    nav {
        background-color: #0f172a;
        padding: 0.75rem 0;
        box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
    }

    nav ul {
        list-style: none;
        display: flex;
        justify-content: center;
        gap: 1.5rem;
    }

    nav ul li a {
        color: #e2e8f0;
        text-decoration: none;
        font-weight: 600;
        padding: 0.5rem 1rem;
        border-radius: 4px;
        transition: all 0.3s ease;
    }

    /* Hover Effect for Links - Light Grey Highlight */
    nav ul li a:hover {
        background-color: #94a3b8;
        color: #0f172a;
    }

    /* Main Layout */
    main {
        max-width: 1100px;
        margin: 2rem auto;
        padding: 0 1rem;
        flex: 1;
    }

    /* Canvas Section - White Card with Slate Grey Border */
    .canvas-section {
        background-color: #ffffff;
        padding: 1.5rem;
        border-radius: 12px;
        box-shadow: 0 4px 6px rgba(0, 0, 0, 0.05);
        margin-bottom: 2.5rem;
        text-align: center;
        border: 1px solid #cbd5e1;
    }

    .canvas-section h2 {
        margin-bottom: 1rem;
        color: #334155;
        font-size: 1.8rem;
    }

    #myCanvas {
        border: 2px solid #475569;
        border-radius: 8px;
        background-color: #f8fafc;
        box-shadow: inset 0 0 5px rgba(0, 0, 0, 0.08);
    }

```

```

/* Content Cards Container */
.cards-container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
  gap: 1.5rem;
  margin-bottom: 2rem;
}

/* Content Card Styling */
.card {
  background-color: #ffffff;
  border-radius: 10px;
  padding: 1.5rem;
  border: 1px solid #cbd5e1;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.05);
  transition: transform 0.3s ease, box-shadow 0.3s ease, border-color 0.3s ease;
}

/* Hover Effect for Cards */
.card:hover {
  transform: translateY(-5px);
  box-shadow: 0 10px 15px rgba(0, 0, 0, 0.1);
  border-color: #64748b;
}

.card h3 {
  color: #1e293b;
  margin-bottom: 0.75rem;
  font-size: 1.4rem;
  text-shadow: 1px 1px 2px rgba(0, 0, 0, 0.05);
}

.card p {
  color: #475569;
  margin-bottom: 1.25rem;
}

/* Section Title Utility */
.section-title {
  text-align: center;
  margin-bottom: 1.5rem;
  color: #1e293b;
}

/* Button Styling - Dark Slate Grey with Smooth Hover */
.card-btn {
  background-color: #475569;
  color: #ffffff;
  border: none;
  padding: 0.6rem 1.2rem;
  font-size: 0.95rem;
  font-weight: 600;
  border-radius: 6px;
  cursor: pointer;
  transition: background-color 0.3s ease, transform 0.2s ease;
}

/* Hover Effect for Buttons */
.card-btn:hover {
  background-color: #1e293b;
  transform: scale(1.05);
}

```

```
/* Footer Styling - Dark Charcoal Header Match */
footer {
  background-color: #0f172a;
  color: #94a3b8;
  text-align: center;
  padding: 1.5rem;
  margin-top: auto;
  border-top: 3px solid #64748b;
}

footer p {
  font-size: 0.9rem;
}
```

7. CODE EXPLANATION

Code Construct / Element	Technical Explanation
linear-gradient(...)	Creates smooth multi-stop color transitions for the body, header, and visual backgrounds.
box-shadow: 0 4px 6px...	Applies elevation and drop shadow to cards and canvas containers for visual depth.
transform: translateY(-5px)	Shifts content cards upward by 5 pixels on hover to simulate tactile floating elevation.
transition: all 0.3s ease	Animates property modifications (transform, shadow, color) over 300 milliseconds.
display: grid (cards-container)	Organizes content cards in a responsive auto-fitting grid layout across screen resolutions.
getContext('2d')	Initializes the 2D rendering context object used to draw vectors and shapes on the canvas surface.
fillRect() / strokeRect()	Draws a solid filled rectangle and outlines its perimeter with coordinates (x, y, width, height).
arc(250, 80, 45, 0, 2*Math.PI)	Traces a circular path specified by center coordinates (250, 80) and radius 45 pixels.
moveTo() / lineTo()	Moves the drawing cursor and constructs straight path vectors connecting coordinate points.
fillText('Abhishek', 250, 200)	Renders styled text 'Abhishek' directly onto the canvas buffer at centered position (250, 200).

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load My-Styled-Webpage.html in browser	Header, navigation, canvas, and cards render with slate theme	Complete page loaded with expected grey/slate gradient aesthetic	Pass
TC-02	Inspect HTML5 Canvas surface	Canvas initializes with 500x250 dimensions and slate border	Canvas rendered with border and inset shadow cleanly	Pass
TC-03	Verify Canvas Rectangle rendering	Blue rectangle (120x80) with dark border rendered at (40, 40)	Rectangle rendered accurately at specified coordinates	Pass
TC-04	Verify Canvas Circle rendering	Red circle with radius 45 rendered centered at (250, 80)	Circle rendered with red fill and dark red stroke outline	Pass
TC-05	Verify Canvas Line rendering	Green diagonal line (lineWidth: 5) connects (350, 40) to (460, 120)	Green vector line drawn cleanly with 5px thickness	Pass
TC-06	Verify Canvas Name text rendering	Student name 'Abhishek' rendered in bold centered at (250, 200)	Name text displayed legibly and centered horizontally	Pass
TC-07	Hover mouse cursor over a Content Card	Card elevates upward (-5px) and shadow deepens smoothly	Smooth translateY transition and shadow expansion observed	Pass
TC-08	Hover mouse cursor over a Card Button	Button background darkens and scales up to 1.05x smoothly	Button hover animation triggered with scale transform	Pass

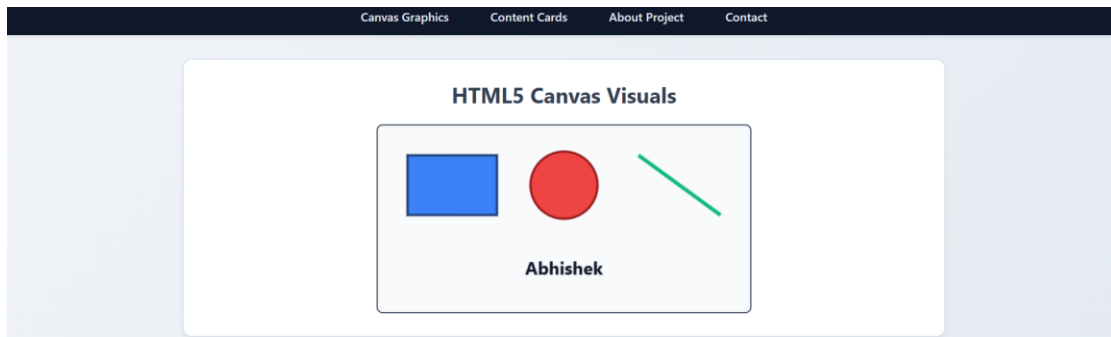
Test Case Execution Screenshots & Verification Evidence:

TC-01: HTML5 Canvas 2D Graphics Execution (Rectangle, Circle, Line, and Text)

Action / Input: Initialize 2D canvas context and render geometric primitives with styled typography.

Expected: Blue rectangle with border, red circle with stroke, green diagonal line, and bold text 'Abhishek' rendered at accurate coordinates.

Observed: All 4 graphical primitives rendered sharply onto the HTML5 canvas element. | Status: **PASS**

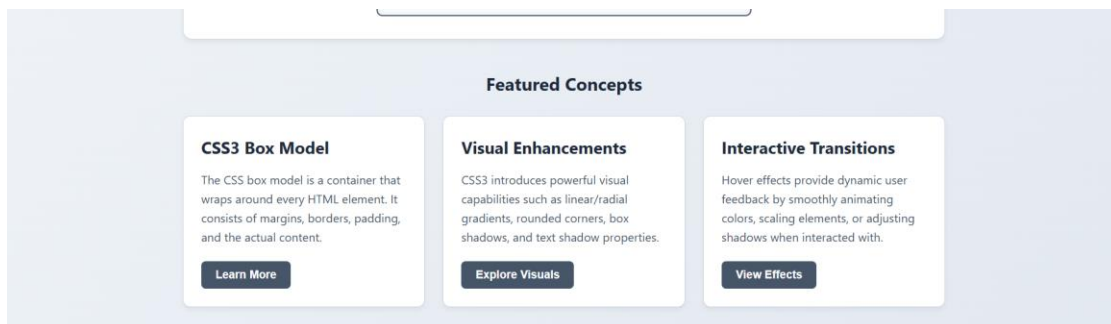


TC-02: CSS3 Box Model & Interactive Content Cards Layout

Action / Input: Inspect featured concepts section displaying responsive CSS3 content cards with grey/slate gradient theme.

Expected: Cards container arranges 3 cards in responsive grid with borders, rounded corners, drop shadows, and dark slate action buttons.

Observed: Responsive grid displayed Box Model, Visual Enhancements, and Interactive Transitions cards. | Status: **PASS**

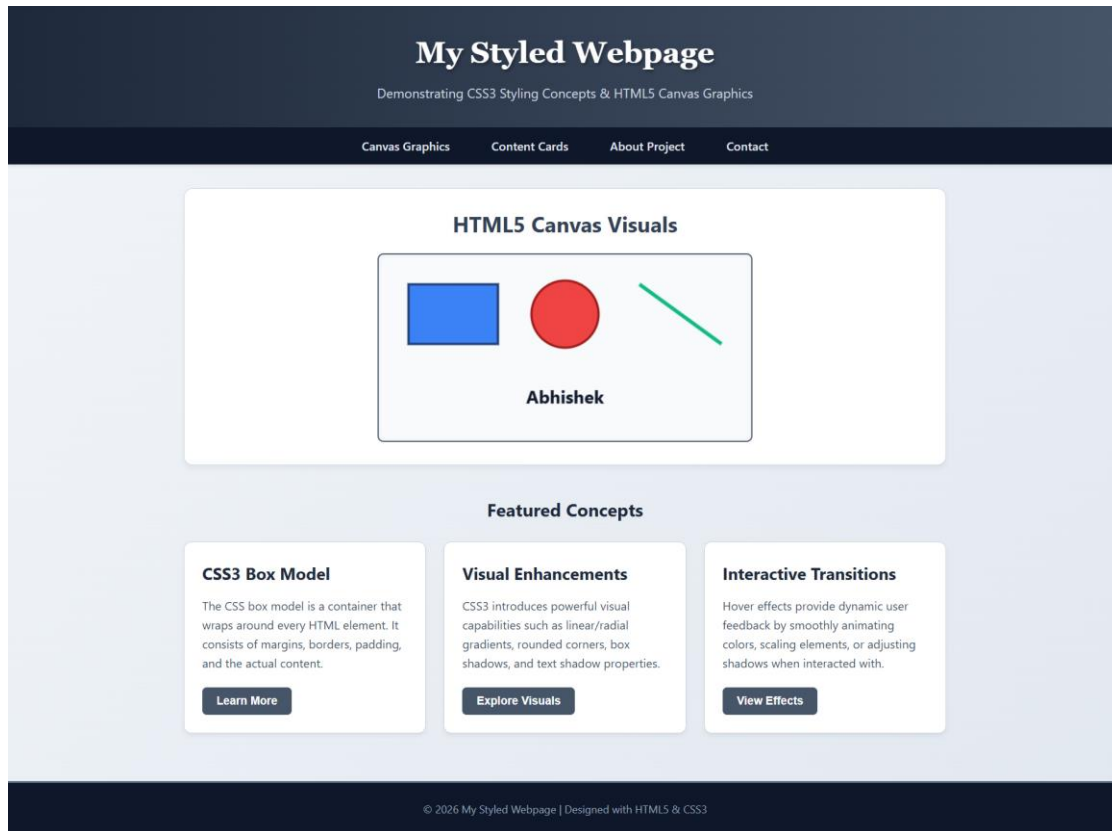


TC-03: Full Webpage Integration & Navigation Header

Action / Input: Load full webpage to evaluate header text shadow, graphite navigation bar, and footer alignment.

Expected: Charcoal gradient header with text shadow, flex navigation bar, and full page layout render cleanly.

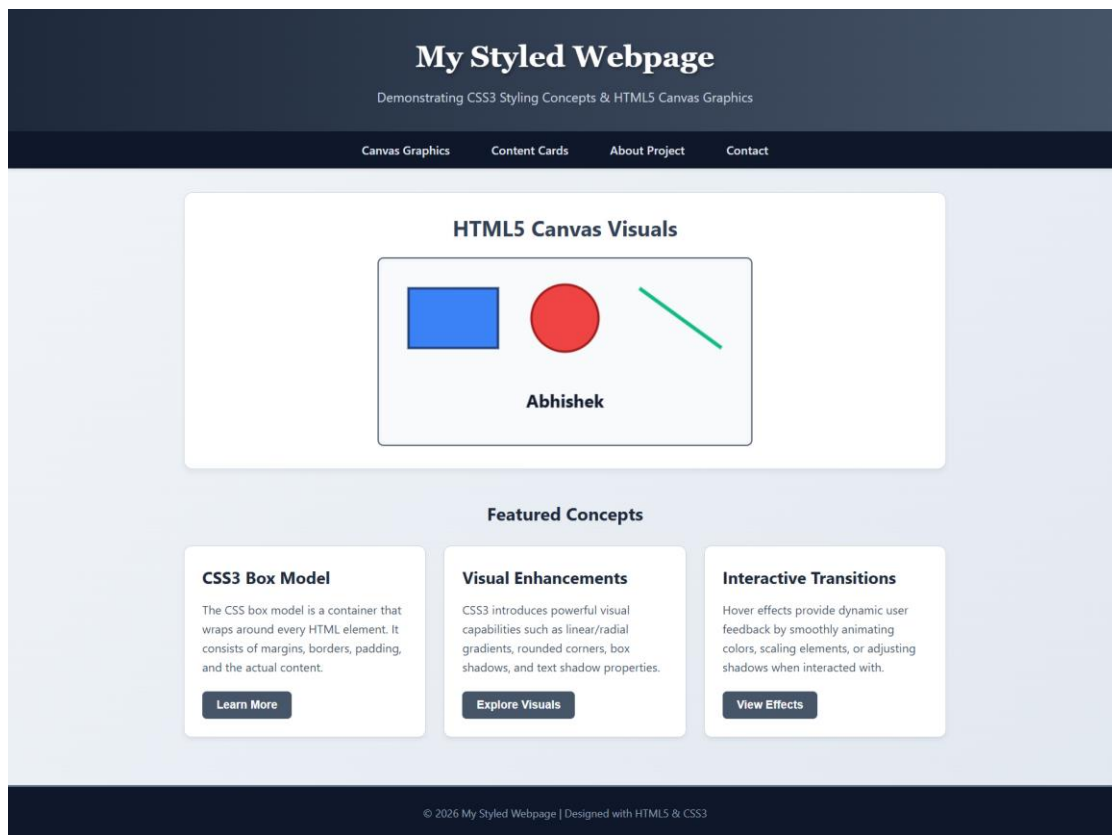
Observed: Complete webpage displayed cohesive slate/grey theme from header to footer. | Status: **PASS**



9. OUTPUT

Output 1: My Styled Webpage with HTML5 Canvas Graphics & Interactive Content Cards

Full browser rendering showing the gradient header, navigation bar, canvas section with four graphical elements (rectangle, circle, line, and student name Abhishek), and the three interactive CSS3 content cards.



10. RESULT

Thus, an interactive webpage demonstrating CSS3 styling concepts (box model, linear gradients, box shadows, transitions, flexbox and grid layouts) and HTML5 Canvas 2D graphics programming was successfully designed, developed, and verified across all test cases with visual execution proof.